Murtaza Amjad

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Personal Website | LinkedIn | GitHub

EDUCATION

University of New Haven, West Haven, CT — May 2026
Bachelors of Science, Computer Science — **GPA:** 3.67/4.00, *Dean's List*

EXPERIENCE

University of New Haven, West Haven, CT — Feb 2026 – May 2026
Software Engineering Research Assistant V & Team Lead

- Built **5+** analytics dashboard using **React** and **Python FastAPI** across the manufacturing sector for **50+** project managers and executives, saving an average of **$20K** in operational costs.
- Integrated **3** disconnected systems (**Sage 100 financials**, **Tekla PowerFab fabrication**, **TSheets time tracking**) into a unified **PostgreSQL** data warehouse, using the shared job number to link records across all three with a **REST API** fetching over **500 GB+** of data.
- Designed and implemented a **3-tier Role-Based Access Control (RBAC)** system (System Admin, Business Admin, Viewer) covering **50+** users.
- Created **3** data pipelines to process, filter, and reorganize data so it stayed in sync and was easily read by the React dashboard.
- Translated **50+ stakeholder requirements** into 15 dashboard features, 1 AI-assisted agent, and 50+ feature/software updates supporting the API and dashboards.
- Implemented a high-fidelity sequence management UI capable of tracking **100+ fabrication sequences** per project.
- Wrote **10+** pages of technical documentation covering system architecture, data schemas, and setup used by the technical team to troubleshoot issues post-deployment.
- Managed the team's GitHub workflow conducting **30+ code reviews**, approved **10+ merges**, and redistributed tasks and workload based on each developer's strength and weaknesses to keep **5+ sprints** on track.
- Led a team of **4** developers, facilitating cross-team discussions and communicating directly with executives and stakeholders.

Projects

SendIt, University of New Haven, CT (Hybrid) — Jan 2025 – Present

- Architected a cross-platform **P2P file transfer system** using **Java, Kotlin, Swift**, and **Spring Boot** supporting secure device-to-device sharing across **iOS**, **Android**, and **desktop**.
- Co-led a team of **6** developers to build a native **iOS** client using SwiftUI and a native **Android** client using Kotlin covering **10+** core screens and navigation.
- Designed and built the sender-side flow and encryption implementation using ECC (P256) key exchange, HKDF (SHA-256) key derivation, and AES-256-GCM encryption.
- Integrated **iOS** and desktop clients with **Spring Boot REST APIs** for account registration and login, secured with **Spring Security** and BCrypt password hashing.
- Led **Agile** development using **Trello** for **5+** sprints, from architecture and **UML** design through implementation, testing, and documentation.
- Directed manual **QA testing** by assigning **2+ team members** to stress-test the app for bugs, logging findings in **Trello** for sprint-based triage, supplemented by in-app error logging to catch runtime failures.
- Presented a public demo in front of a panel of **5+** judges and industry professionals and made the **final round** at the **Tagliatela College of Engineering Capstone Expo.**

PokéPlatform, University of New Haven, CT — Nov 2025 – Dec 2025

- Built a deployed full-stack web application (**Vue.js, Express.js, MongoDB**) with **REST API's** supporting full **CRUD** functionality allowing me to update the Pokédex.
- Manually tested and validated **4+ REST API** endpoints using **Postman** throughout development.

SKILLS

Programming: Java, C++, JavaScript, TypeScript, Python, C#, Swift, Kotlin, SQL, C, HTML, CSS
Frameworks & Libraries: Spring Boot, React, Vue/Express/Node.js
Tools & Practices & Certifications: AWS, Git/GitHub Actions, Figma, Linux, Docker, Postman, Unity, Agile, JUnit Testing, ***Lean Six Sigma Yellow Belt Certificate***

ACTIVITIES AND LEADERSHIP

Computer Programming Club (*President),* **Upsilon Pi Epsilon** (*President)*, **NorthEast Annual Cybersecurity Summit** (*Student Ambassador*), **Wesleyan University Hackathon** (Front End Lead)